2.10.2 Relative Velocity in One and Two Dimensional Motion

When two objects A and B are moving with different velocities, then the velocity of one object A with respect to another object B is called relative velocity of object A with respect to B.

Case 1

Consider two objects A and B moving with uniform velocities V_A and $V_{B,}$ as shown, along straight tracks in the same direction $\vec{V}_A$, $\vec{V}_B$ with respect to ground.

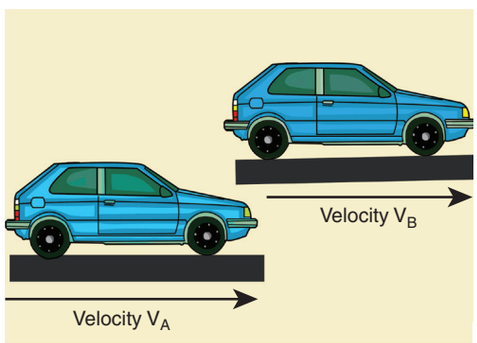
The relative velocity of object A with respect to object B is $\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$

The relative velocity of object B with respect to object A is $\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$

Thus, if two objects are moving in the same direction, the magnitude of relative velocity of one object with respect to another is equal to the difference in magnitude of two velocities.

EXAMPLE 2.26

Suppose two cars A and B are moving with uniform velocities with respect to ground along parallel tracks and in the same direction. Let the velocities of A and B be 35 km h^{-1} due east and 40 km h^{-1} due east respectively. What is the relative velocity of car B with respect to A?



Solution

The relative velocity of B with respect to A, $\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = 5$ km h^{-1} due east

Similarly, the relative velocity of A with respect to B i.e., $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = 5$ km h^{-1} due west.

To a passenger in the car A, the car B will appear to be moving east with a velocity 5 km h^{-1}. To a passenger in car B, the car A will appear to move westwards with a velocity of 5 km h^{-1}

Case 2

Consider two objects A and B moving with uniform velocities V_A and V_B along the same straight tracks but opposite in direction



The relative velocity of object A *with respect to* object B is

$$\vec{V}_{AB} = \vec{V}_A - (-\vec{V}_B) = \vec{V}_A + \vec{V}_B$$

The relative velocity of object B with respect to object A is

$$\vec{V}_{BA} = -\vec{V}_B - \vec{V}_A = -(\vec{V}_A + \vec{V}_B)$$

Thus, if two objects are moving in opposite directions, the magnitude of relative velocity of one object with respect to other is equal to the sum of magnitude of their velocities.

Case 3

Consider the velocities $\vec{v}_A$ and $\vec{v}_B$ at an angle θ between their directions.

The relative velocity of A with respect to B, $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$

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